School Worksheet

Class 12 Mathematics, Chapter 3: Matrices

Maximum marks: **49** **20** questions

Name _____ Date _____ Score _____ / 49

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

Let $A = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 4 & -1 \\ -3 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -2 \\ -5 \\ -7 \end{bmatrix}$, $C = \begin{bmatrix} 9 & 8 & 7 \end{bmatrix}$. Which of the following is defined?

(A) Only AB **(B)** Only AC

(C) Only BA **(D)** All AB, AC and BA

2 MCQ — 1 mark

If $\begin{bmatrix} x+y & 3y \\ 3x & x+3 \end{bmatrix} = \begin{bmatrix} 9 & 4x+y \\ x+6 & y \end{bmatrix}$, then $(x-y)$ is equal to:

(A) -7 **(B)** -3 **(C)** 3 **(D)** 7

3 MCQ — 1 mark

If $A = [a_{ij}]$ is a skew-symmetric matrix of order n, then a_{ii} is equal to:

(A) 1 **(B)** 0 **(C)** -1 **(D)** n

4 MCQ — 1 mark

Let $A = [a_{ij}]$ be a 2×2 matrix whose elements are given by $a_{ij} = \dfrac{(i+2j)^2}{3}$. Then A' is:

(A) $\begin{bmatrix} 12 & \frac{25}{3} \\ \frac{16}{3} & 3 \end{bmatrix}$ **(B)** $\begin{bmatrix} 12 & \frac{16}{3} \\ \frac{25}{3} & 3 \end{bmatrix}$ **(C)** $\begin{bmatrix} 3 & \frac{25}{3} \\ \frac{16}{3} & 12 \end{bmatrix}$ **(D)** $\begin{bmatrix} 3 & \frac{16}{3} \\ \frac{25}{3} & 12 \end{bmatrix}$

5 MCQ — 1 mark

If A is a square matrix such that $A^2 = A$, then $(I+A)^3 - 7A$ is equal to:

(A) A **(B)** $I - A$ **(C)** I **(D)** $3A$

Q No.	Question	Marks
6	Assertion-Reason **Assertion (A):** If A and B are two square matrices such that AB and BA are defined, then it is not necessary that $AB = BA$. **Reason (R):** The product of two diagonal matrices of the same order is commutative. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer Find x and y if $2\begin{bmatrix}1 & 3\\0 & x\end{bmatrix} + \begin{bmatrix}y & 0\\1 & 2\end{bmatrix} = \begin{bmatrix}5 & 6\\1 & 8\end{bmatrix}$.	2 marks

Level 2

18 marks

8	Short answer Construct a 3×2 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = \frac{1}{2}\lvert i - 3j\rvert$.	2 marks
9	Short answer If $A = \begin{bmatrix}1 & 2\\2 & 1\end{bmatrix}$, show that $A^2 - 2A - 3I = O$. Hence find A^{-1}.	3 marks
10	Short answer Express $A = \begin{bmatrix}3 & 5\\1 & -1\end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.	3 marks
11	Short answer If $A = \begin{bmatrix}8 & 0\\4 & -2\\3 & 6\end{bmatrix}$ and $B = \begin{bmatrix}2 & -2\\4 & 2\\-5 & 1\end{bmatrix}$, find the matrix X such that $2A + 3X = 5B$.	2 marks
12	Short answer Find x if $\begin{bmatrix}x & -5 & -1\end{bmatrix}\begin{bmatrix}1 & 0 & 2\\0 & 2 & 1\\2 & 0 & 3\end{bmatrix}\begin{bmatrix}x\\4\\1\end{bmatrix} = O$.	3 marks
13	Short answer If $A = \begin{bmatrix}\cos\alpha & -\sin\alpha\\\sin\alpha & \cos\alpha\end{bmatrix}$, find $\alpha\left(0 < \alpha < \frac{\pi}{2}\right)$ such that $A + A' = I$.	2 marks
14	Short answer If $A = \begin{bmatrix}1\\-4\\3\end{bmatrix}$ and $B = \begin{bmatrix}-1 & 2 & 1\end{bmatrix}$, verify that $(AB)' = B'A'$.	3 marks

Q No.	Question	Marks

Level 3 — 23 marks

15 Long answer — 3 marks

If $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$, find $A^2 - 5A + 4I$ and hence find a matrix X such that $A^2 - 5A + 4I + X = O$.

16 Long answer — 3 marks

A trust fund has ₹30,000 to be invested in two types of bonds paying 5% and 7% interest per annum. Using matrix multiplication, determine how to divide ₹30,000 among the two bonds so that the trust fund earns an annual interest of ₹1,800.

17 Long answer — 3 marks

If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, show that $A^2 = 3A$ and hence find A^3.

18 Long answer — 5 marks

If $A = \begin{bmatrix} 0 & -\tan\frac{\alpha}{2} \\ \tan\frac{\alpha}{2} & 0 \end{bmatrix}$ and I is the identity matrix of order 2, show that $I + A = (I - A)\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$.

19 Long answer — 5 marks

If $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix}$, show that $A^3 - 23A - 40I = O$.

20 Case study — 4 marks

Case Study: Three schools P, Q and R organised a fair to raise funds by selling hand-made fans, mats and plates made from recycled material at ₹25, ₹100 and ₹50 each respectively. School P sold 40 fans, 50 mats and 20 plates; School Q sold 25 fans, 40 mats and 30 plates; School R sold 35 fans, 50 mats and 40 plates.

(i) Write the sales of the three schools as a 3×3 matrix and the prices as a column matrix. — 1 mark

(ii) Find the funds collected by School P. — 1 mark

(iii) Using matrix multiplication, find the funds collected by each school and the total funds collected. — 2 marks

Answer key

Chapter 3: Matrices, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) Only AB
2	(B) -3
3	(B) 0
4	(D) $\begin{bmatrix} 3 & \frac{16}{3} \\ \frac{25}{3} & 12 \end{bmatrix}$
5	(C) I
6	(B) Both A and R are true, but R is not the correct explanation of A.
7	$x = 3,\ y = 3$

Level 2

8	$A = \begin{bmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{bmatrix}$
9	$A^{-1} = \frac{1}{3}(A - 2I) = \frac{1}{3}\begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$
10	$A = \begin{bmatrix} 3 & 3 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$
11	$X = \begin{bmatrix} -2 & -\frac{10}{3} \\ 4 & \frac{14}{3} \\ -\frac{31}{3} & -\frac{7}{3} \end{bmatrix}$
12	$x = \pm 4\sqrt{3}$
13	$\alpha = \frac{\pi}{3}$
14	$(AB)' = B'A' = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$, hence verified.

Level 3

15	$A^2 - 5A + 4I = \begin{bmatrix} -1 & -1 & -3 \\ -1 & -3 & -10 \\ -5 & 4 & 2 \end{bmatrix}$; $X = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 10 \\ 5 & -4 & -2 \end{bmatrix}$
16	₹15,000 in the 5% bond and ₹15,000 in the 7% bond.
17	$A^3 = 9A = \begin{bmatrix} 9 & 9 & 9 \\ 9 & 9 & 9 \\ 9 & 9 & 9 \end{bmatrix}$
18	RHS simplifies to $\begin{bmatrix} 1 & -\tan\frac{\alpha}{2} \\ \tan\frac{\alpha}{2} & 1 \end{bmatrix} = I + A$. Hence proved.
19	$A^3 = \begin{bmatrix} 63 & 46 & 69 \\ 69 & -6 & 23 \\ 92 & 46 & 63 \end{bmatrix}$ and $A^3 - 23A - 40I = O$. Hence proved.
20	P: ₹7,000; Q: ₹6,125; R: ₹7,875; total funds ₹21,000.